

# Hongchen Wei

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**Agentic AI | Generative Environments | Post-training**

## Education

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| <b>Wuhan University</b> Ph.D. student; advisor: Prof. Zhenzhong Chen (IIP Lab) | Sep. 2023 – Jun. 2027 |
| Nanjing University of Science and Technology M.S.                              | Sep. 2020 – Apr. 2023 |
| Xi'an Shiyou University B.E.                                                   | Sep. 2016 – Jun. 2020 |

## Research Overview

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**Research agenda:** Generalist agents for professional work requiring information integration, tool use, and concrete deliverables, with a focus on generative environments, synthetic data, post-training, and verifiable evaluation.

**Current focus:** Leading OfficeTown with collaborators to generate tasks, working materials, and feedback through ongoing organizational simulation, providing scalable training and evaluation resources.

**Methods and evaluation:** Mutable-state interaction, active memory, in-environment RL, capability transfer, and self-alignment; evidence-level diagnosis of capability limits and failure modes.

## Research Experience

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| <b>Microsoft Research Asia (MSRA) Visual Computing Group</b> | Beijing, China             |
| <i>Intern; Mentor: Dr. Bei Liu</i>                           | <i>Dec. 2025 – Present</i> |

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| <b>OfficeTown</b>   <i>Generative Organizations</i> | <i>ICLR 2027 (under review)</i> |
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- Led **OfficeTown** with collaborators to develop a generative organizational simulation framework that produces professional tasks, supporting materials, and feedback through ongoing organizational activity, providing data and executable environments for agent evaluation, supervised fine-tuning, and RL.
- Generated organizations with professional roles, shared knowledge, customer relationships, and tool-enabled workspaces from natural-language business briefs. Built **23 organizations across 8 professional domains** and a benchmark with expert-reviewed tasks and rubrics; evaluated frontier models to identify capability gaps. Training **Qwen3.5-9B** on synthetic data substantially improved **GDPval** and **Jobbench** performance. The principal–assistant harness outperformed the single- and multi-agent baselines tested.

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| <b>DocAtlas</b>   <i>Mutable-State Document Interaction</i> | <i>NeurIPS 2026</i> |
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- Led **DocAtlas** with collaborators, reframing long-document understanding as mutable-state interaction rather than static retrieval-based QA. Designed a trainable harness with Search, Read, Note, and Review, self-improving retrieval, and active memory for large-model inference and RL.
- On MMLongBench-Doc, GPT-5.4 + DocAtlas reached **71.4%** (human-expert reference: 65.8%). With in-environment RL, Qwen3.5-4B reached **63.7%**, versus a 54.4% baseline (**+9.3 percentage points**).

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| <b>XL-DocBench</b>   <i>Evidence-Level Evaluation</i> | <i>NeurIPS 2026</i> |
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- Led **XL-DocBench** with collaborators, combining expert-annotated evidence pages, typed verification rules, and **12 reasoning labels** for auditable diagnosis of retrieval, evidence-use, and rule-execution errors beyond final-answer scoring.
- Validated by **194 domain experts: 1,519 questions across 6 domains**, with contexts up to **2,303 pages**, including multi-page, cross-document, multimodal, and unanswerable questions. Further synthesized approximately **40K** post-training examples.

**PASA / ANO** | *Capability Transfer and Self-Alignment*

- Led **PASA** with collaborators, characterizing post-merge perception–reasoning asymmetry and developing detached-anchor asymmetric GRPO for self-alignment, recovering visual perception and improving reasoning without ground-truth answer labels.
- Led **ANO** with collaborators, developing training-free, layer-specialized model merging guided by visual-attention differences. Transferred language-model reasoning, coding, and GUI capabilities while preserving visual perception, without additional multimodal training data or gradient updates.

**Long-Context Understanding and Generation**

- Led studies with collaborators on **Visual Context Window Extension** for long-video understanding and **RealVG** for open-world spatio-temporal grounding, extending to long-caption generation and unsupervised prompt learning.

**Selected Publications**

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**18 papers** (published or under review): **11 first-author** and 7 co-authored. Selected papers below.

**First Author: Under Review**

- **Hongchen Wei**, et al. “OfficeTown: Generative Organizations for Evolving Agent Intelligence.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “PASA: Post-Merge Perception–Reasoning Asymmetry as a Self-Alignment Signal for MLLMs.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “Layer-Specialized Model Merging for Training-Free Cross-Modal Reasoning Transfer (ANO).” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “LongCaption: Unlocking the Power of Long Caption Generation in Large Multimodal Models.” *ACM TIST (under review)*.

**First Author: Published**

- **Hongchen Wei**, et al. “DocAtlas: Long-Document Understanding as Mutable-State Interaction.” *NeurIPS 2026*.
- **Hongchen Wei**, et al. “XL-DocBench: Benchmarking Evidence-Grounded Extra-Long Document Understanding.” *NeurIPS 2026*.
- **Hongchen Wei**, et al. “Visual Context Window Extension: A New Perspective for Long Video Understanding.” *ACM MM 2025*.
- **Hongchen Wei**, et al. “RealVG: Unleashing MLLMs for Training-Free Spatio-Temporal Video Grounding in the Wild.” *ACM MM 2025*.
- **Hongchen Wei**, et al. “Improving Generalization of Image Captioning with Unsupervised Prompt Learning.” *ACM TOMM 2024*.
- **Hongchen Wei**, et al. “Exploiting Cross-Modal Prediction and Relation Consistency for Semi-Supervised Image Captioning.” *IEEE TCYB 2023*.
- **Hongchen Wei**, et al. “S2OSC: A Holistic Semi-Supervised Approach for Open Set Classification.” *ACM TKDD 2022*.

**Selected Co-Authored Papers**

- “See What We Cannot See: A Geo-guided Reasoning Benchmark for Object Counting under Adverse Earth Observation Conditions.” *CVPR 2026*.
- “TDSAgent: A Task-Driven Sampling Agent for Long Video Question Answering.” *IEEE TMM 2026*.
- “Remote Sensing Semantic Segmentation Quality Assessment based on Vision Language Model.” *IEEE TGRS 2025*.